Automatic Grid-Following ↔ Grid-Forming Mode Switching of One Synchronous Generator and Four IBRs on the IEEE 14-Bus System via AGSI++

Small-signal stability sweep, fault switching-outcome map, and time-domain trip/reclose switching (project-owned MATLAB, no external solver)

3 August 2026

Abstract

This standalone report consolidates the IEEE 14-bus study of automatic grid-following (GFL) ↔ grid-forming (GFM) mode switching driven by the Adaptive Grid-Support Index (AGSI++). The test system has **one synchronous generator** at bus 1 (Padiyar model-1.1, *manual field, no AVR*, 4-state) sharing a *meshed single-area* network with **four GFL IBRs** at buses 2, 3, 6 and 8 (reduced-6 `SwitchableIbr6`). The device models are identical to those of the companion Padiyar two-area switching study [1]. The composite small-signal model has **28 states** and **zero right-half-plane eigenvalues** (max $\mathrm{Re}(\lambda) \approx 0$); the least-damped mode is $f = 0.345$ Hz with $\zeta = +0.267$, so the operating point is *small-signal stable* and satisfies the no-AVR / no-PSS, 1-SG + 4-IBR constraint. The report brings together (A) an SSSA stability sweep over inertia and internal-reactance (coupling) scaling, (B) a fault switching-outcome map over fault severity and location, and (C) the power-flow operating point, the SSSA verdict, the time-domain SG trip/reclose switching cycle, and the AGSI++ weight-tuning provenance. All production dynamics are project-owned base MATLAB (audited `lu`/`eig` primitives only); external programs are never numerical acceptance targets.

1 Scope and link to the companion report

This report is standalone but reuses, as the single source of truth, the reduced-6 IBR models (GFL, GFM), the `SwitchableIbr6` bumpless device, the Padiyar model-1.1 synchronous machine, and the AGSI++ supervisor documented and derived in full in the companion report `report_two_ibr_switch_en` [1, 2, 3, 4]. Here those models are applied to the meshed IEEE 14-bus system, and three connected artifacts are added: a small-signal stability sweep (§5), a fault switching-outcome map (§7), and the time-domain trip/reclose switching cycle (§6). Source discipline: the AGSI++ structure follows the EECON49-P4 strategy [1] as a design guideline (`PROJECT_DERIVED_SOURCE_GUIDED`); the diagnostic time-domain driver and the two sweeps are `ASSUMED_DIAGNOSTIC`; nothing was tuned to bias any result.

2 Models and the AGSI++ switching index

Switching index. The AGSI++ merges six normalised per-IBR sub-indices into one dimensionless index, evaluated every step:

$$\begin{aligned} J_V &= \frac{|V - V_{ref}|}{0.10}, \quad J_f = \frac{|f - 60|}{0.50}, \quad J_R = \frac{|\mathrm{d}f/\mathrm{d}t|}{1.0}, \\ J_P &= \frac{|P_{ref} - P|}{0.20}, \quad J_{SCR} = \max\Big(0, \tfrac{3}{SCR} - 1\Big), \quad J_{lock} = \frac{|v_q|}{0.10}, \end{aligned} \tag{1}$$

$$\mathrm{AGSI} = w_V J_V + w_f J_f + w_R J_R + w_P J_P + w_{SCR} J_{SCR} + w_{lock} J_{lock}, \tag{2}$$

V is the local PCC voltage magnitude ($V_{ref} = 1$ pu); f is PLL-estimated frequency in GFL and virtual-rotor frequency in GFM ($f_0 = 60$ Hz); $\mathrm{d}f/\mathrm{d}t$ is the accepted-sample backward-difference RoCoF, low-pass filtered in AGSI++; P is injected active power on the system base with generator-positive sign and P_{ref} its command; SCR is the diagnostic local short-circuit ratio; and v_q is the active-frame q-axis voltage (PLL frame in GFL, virtual-rotor frame in GFM). The denominators are frozen normalisation bases, not protection limits: $J_V = 1$, for example, means a 0.10 pu voltage deviation. Sub-indices and the aggregate are dimensionless, are not probabilities, are not clipped to $[0, 1]$, and may exceed one. with weights (sum = 1) $[w_V, w_f, w_R, w_P, w_{SCR}, w_{lock}] = [0.25, 0.25, 0.15, 0.10, 0.15, 0.10]$. Switching uses hysteresis and dwell: GFL→GFM when AGSI $\geq \Gamma_{on} = 0.65$ (held $\geq T_{d,on}$); GFM→GFL when AGSI $< \Gamma_{off} = 0.35$ (held $\geq T_{d,off}$). AGSI++ low-pass filters the RoCoF term ($\tau_R = 0.05$ s) to reject the one-sample RoCoF spike at a topology change, and adds the weak-grid strength term J_{SCR} and the PLL loss-of-lock proxy $J_{lock} = |v_q|/0.10$.

Code-level switching mechanism. Each IBR is an independent hybrid two-state machine with committed mode $m_i \in \{0, 1\}$ (0 =GFL, 1 =GFM), not an online optimiser or a continuous blend of the two models. Its own AGSI++ and dwell timers determine the normal transitions, so devices switch at the same simulated time only when their local conditions mature together. The IEEE14 run has `gra_override=false`: fault/trip transitions are purely index-driven. A synchronised SG reclose is the explicit exception—the event driver performs a coordinated reference handback to GFL without waiting for Γ_{off}; the index supervisor then resumes immediately.

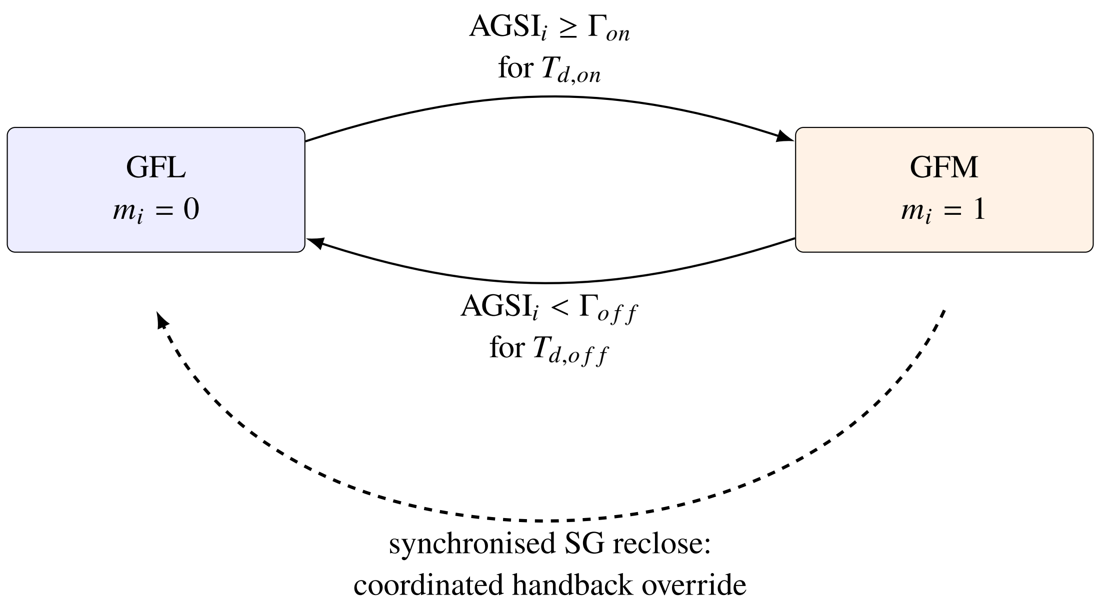

Figure 1: Per-device hybrid state machine. Solid transitions are AGSI++/hysteresis/dwell decisions; the dashed transition is the separately identified SG-reference handback policy.

Reduced-6 IBR models. **GFL** state order $x = [\, i_d,\ i_q,\ \delta_{\text{PLL}},\ \xi_{\text{PLL}},\ \xi_P,\ \xi_Q \,]$ (L-filter dq currents, SRF-PLL angle and integrator, and P/Q outer-loop integrators). **GFM** state order $x = [\, i_d,\ i_q,\ \omega,\ \delta,\ E,\ \xi_V \,]$ (currents, virtual-rotor speed and angle, internal voltage magnitude, and d-axis voltage integrator), with cross-paired terminals $v_{gd} = E + Xi_q$, $v_{gq} = -Xi_d$. Each IBR is one `SwitchableIbr6` device wrapping both branches; the active state is always 6-dimensional and a switch is *bumpless* — the target branch is re-initialised to the instantaneous terminal (V, P, Q) so the injected current is continuous,

$$I^+ = \overline{\left(\frac{S}{V}\right)} = I^-. \tag{3}$$

Synchronous generator (Padiyar model-1.1, manual, 4-state). State order $[\delta,\ \omega,\ E'_q,\ E'_d]$:

$$\dot{\delta} = \omega_B(\omega - 1), \qquad \dot{\omega} = \frac{P_m - (\omega - 1)/R - T_e - D(\omega - 1)}{2H}, \tag{4}$$

with $E_{fd} = E_{fd0}$ constant (no AVR), $\dot{E}'_q = [E_{fd} - E'_q - (X_d - X'_d)I_d]/T'_{d0}$ and $\dot{E}'_d = [-E'_d + (X_q - X'_q)I_q]/T'_{q0}$. The manual (no-AVR, no-PSS) machine keeps the composite free of any excitation-control state, as required by the study constraint. Full derivations, the stator solution, and the network DAE are in the companion report.

3 Test system

Figure 2 is the single-line diagram of the study system: the IEEE 14-bus network with the synchronous generator **SG**$_1$ at bus 1 acting as slack (angle reference) and four inverter-based resources **IBR**$_1$–**IBR**$_4$ at buses 2, 3, 6 and 8, each backed by an energy source and/or storage. In steady state every IBR is dispatched as a PQ (grid-following) unit; when the AGSI++ index commands a switch, the affected unit becomes a voltage-forming (PV / grid-forming) source. The remaining buses carry the standard IEEE 14-bus loads, and the transformer taps of branches 4–7, 4–9 and 5–6 are represented exactly by the tap-aware admittance matrix taken from the in-house power flow. This is the same one-line arrangement used in the EECON49-P4 study [1], which is why the device mapping (1 SG + 4 IBR at the original generator buses) is kept unchanged here.

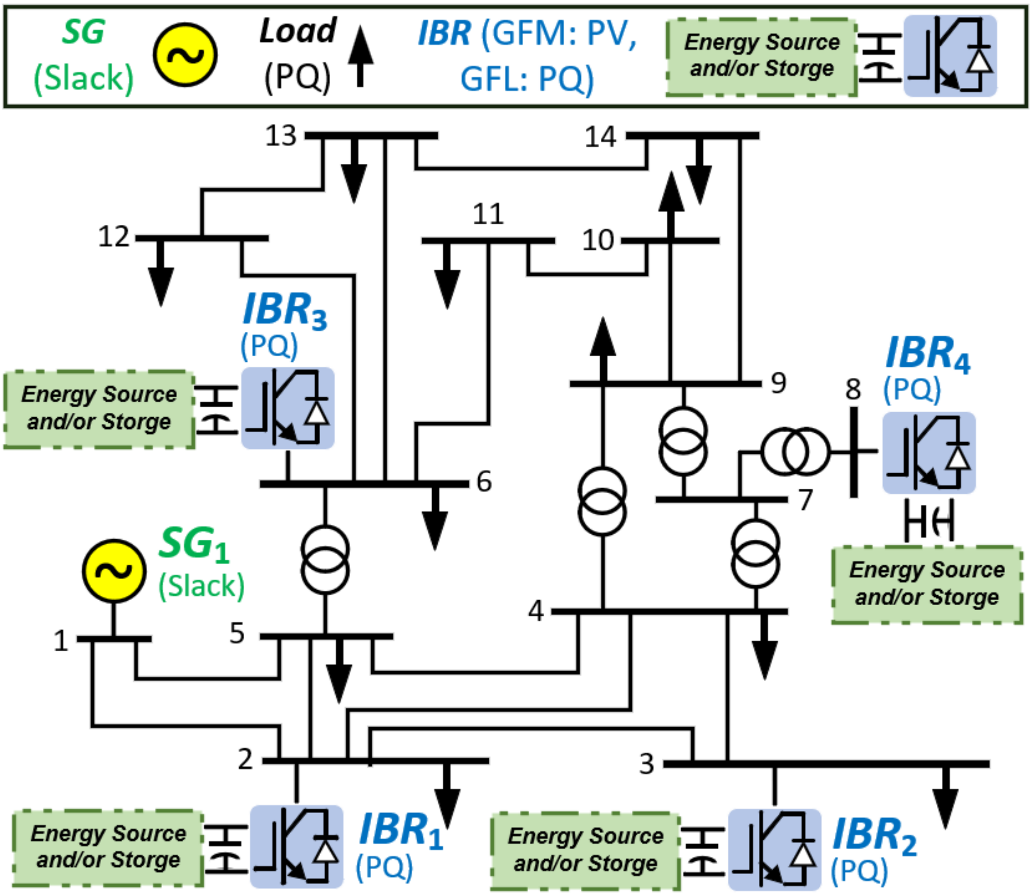

Figure 2: Single-line diagram of the IEEE 14-bus test system used in this study: SG_1 (slack) at bus 1 and the four switchable IBRs at buses 2, 3, 6 and 8 (GFM: PV, GFL: PQ), each with an energy source and/or storage; arrows are loads (PQ). Reproduced from the EECON49-P4 study system [1].

4 Power-flow operating point

The base case is solved by the in-house Newton–Raphson power flow. The synchronous generator at bus 1 carries the bulk dispatch, $P_{g1} = 2.324$ pu $= 232.4$ MW (slack / angle reference). The IBR at bus 2 dispatches $P_g = 0.40$ pu; the IBRs at buses 3, 6 and 8 are on voltage/reactive support duty with scheduled $P_g = 0$. Load is distributed over the remaining IEEE 14-bus buses. This operating point is the initial condition for both the small-signal linearisation (§5) and every time-domain run (§6); the pre-event active-power panel (Fig. 8) shows the SG holding ≈ 2.32 pu and IBR1 holding ≈ 0.40 pu, confirming the dispatch.

5 Small-signal stability and the stability sweep

Linearising the composite 28-state model about the base case gives **zero right-half-plane eigenvalues**: $\max \mathrm{Re}(\lambda) \approx 0$ (only the ≈ 0 angle-reference gauge mode, at the 10^{-8} level), and the least-damped oscillatory mode is $f = 0.345$ Hz with $\zeta = +0.267$. The meshed single-area IEEE 14-bus operating point is therefore *small-signal stable*.

To test how robust this verdict is, two one-parameter sweeps re-derive the equilibrium consistently and re-linearise at each point (Fig. 3, Table 1): an *inertia* sweep on $H_\times$ (scaling SG H and damping) and a *coupling* sweep on k_X (scaling SG internal reactances; larger = weaker coupling). The inertia sweep is essentially flat and stable across $H_\times \in [0.02, 2]$ (max Re $\sim 10^{-8}$, $\zeta \approx 0.267$, mode near 0.345 Hz). The coupling sweep stays stable across $k_X \in [1, 20]$: damping erodes *monotonically* ($\zeta : 0.267 \to 0.150$ at 20× internal reactance) but never crosses $\zeta = 0$, and max Re stays at the 10^{-8} gauge level. **No instability boundary is found on either knob.**

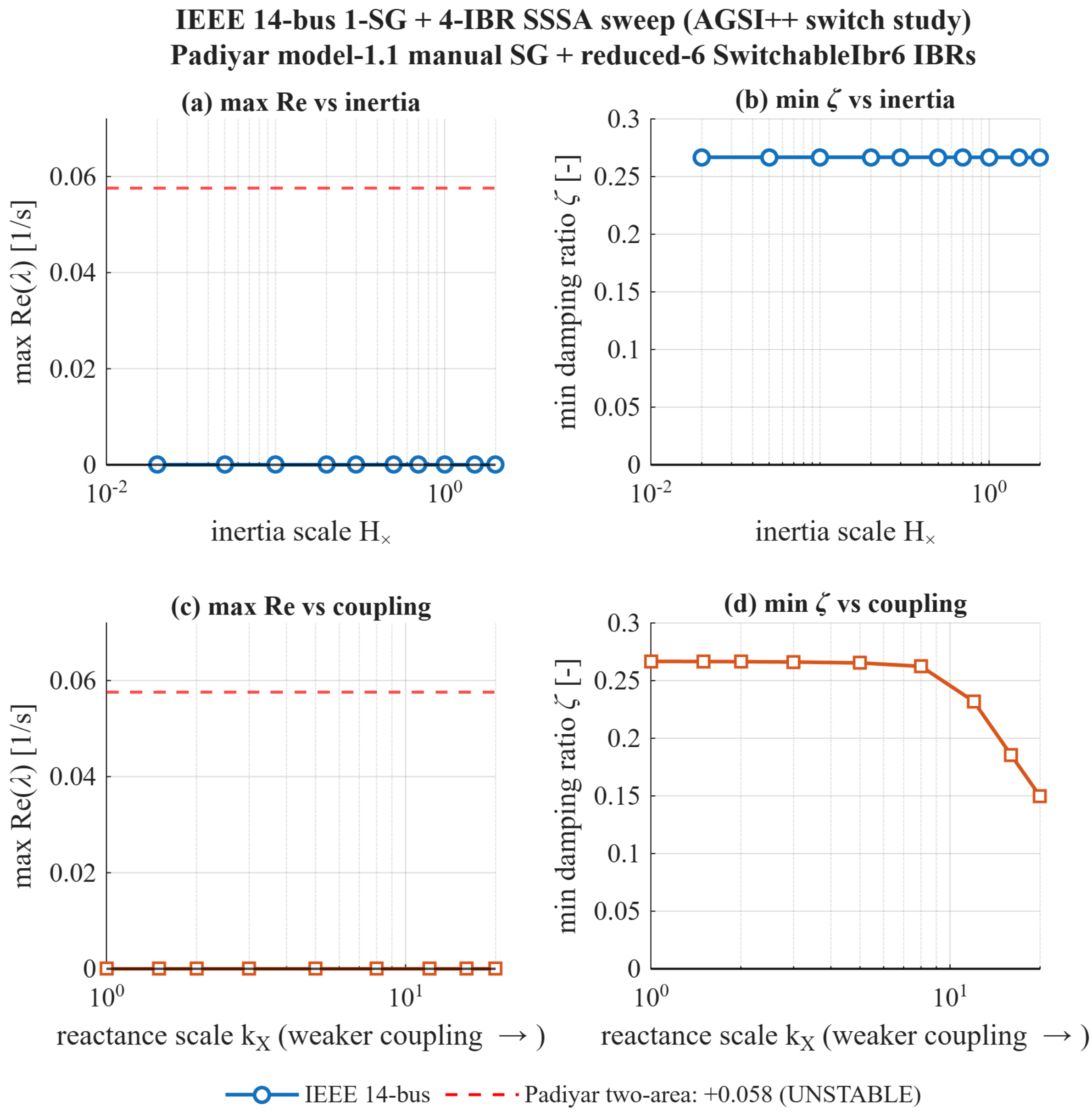

Figure 3: SSSA stability sweep of the IEEE 14-bus 1-SG + 4-IBR system: spectral abscissa max $\mathrm{Re}(\lambda)$ and least-damped ζ versus inertia scaling (top, $H_\times$) and coupling/internal-reactance scaling (bottom, k_X). The IEEE 14-bus system stays stable throughout (max Re $\approx$ 0); the red dashed line marks the Padiyar two-area contrast at max Re = +0.058 s^{-1} (UNSTABLE, a different weak-tie topology).

Table 1: SSSA stability sweep (inertia and coupling knobs) with the Padiyar two-area contrast. Generated entirely by `scripts/reporting/ieee14_sssa_sweep.m`.

Sweep	Param	max Re(λ) (1/s)	$\zeta_{\min}$ (–)	f (Hz)	Verdict (n_{unstable})
IEEE14 inertia sweep ($H_\times$ scales SG H and D; $k_X = 1$)					
$H_\times = 2.00$	2	$+3.222 \times 10^{-8}$	0.2665	0.3448	stable (0)
$H_\times = 1.50$	1.5	$+3.735 \times 10^{-8}$	0.2666	0.3448	stable (0)
$H_\times = 1.00$	1	$+4.448 \times 10^{-8}$	0.2667	0.3449	stable (0)
$H_\times = 0.70$	0.7	$+5.021 \times 10^{-8}$	0.2667	0.3449	stable (0)
$H_\times = 0.50$	0.5	$+5.492 \times 10^{-8}$	0.2667	0.3449	stable (0)
$H_\times = 0.30$	0.3	$+6.065 \times 10^{-8}$	0.2668	0.3449	stable (0)
$H_\times = 0.20$	0.2	$+6.396 \times 10^{-8}$	0.2668	0.3450	stable (0)
$H_\times = 0.10$	0.1	$+6.766 \times 10^{-8}$	0.2668	0.3450	stable (0)
$H_\times = 0.05$	0.05	$+6.971 \times 10^{-8}$	0.2668	0.3450	stable (0)
$H_\times = 0.02$	0.02	$+7.095 \times 10^{-8}$	0.2668	0.3450	stable (0)
IEEE14 grid-coupling sweep (k_X scales X_d, X'_d, X_q, X'_q; $H_\times = 1$)					
$k_X = 1.00$	1	$+4.448 \times 10^{-8}$	0.2667	0.3449	stable (0)
$k_X = 1.50$	1.5	$+5.836 \times 10^{-8}$	0.2666	0.3448	stable (0)
$k_X = 2.00$	2	$+6.357 \times 10^{-8}$	0.2664	0.3447	stable (0)
$k_X = 3.00$	3	$+6.290 \times 10^{-8}$	0.2662	0.3445	stable (0)
$k_X = 5.00$	5	$+5.070 \times 10^{-8}$	0.2654	0.3440	stable (0)
$k_X = 8.00$	8	$+3.453 \times 10^{-8}$	0.2624	0.3425	stable (0)
$k_X = 12.00$	12	$+2.004 \times 10^{-8}$	0.2320	0.3325	stable (0)
$k_X = 16.00$	16	$+7.427 \times 10^{-9}$	0.1855	0.3193	stable (0)
$k_X = 20.00$	20	$+1.179 \times 10^{-8}$	0.1497	0.3117	stable (0)
Contrast: Padiyar two-area 1-SG+3-IBR (weak-tie topology)					
Padiyar	—	$+5.758 \times 10^{-2}$	0.1224	0.6766	**UNSTABLE** (1)

Interpretation (topology, not inertia or coupling). The contrast with the Padiyar two-area system (1 SG +3 IBR, weak inter-area tie) is decisive: that operating point has one right-half-plane eigenvalue, max Re $= +0.058\ \text{s}^{-1}$ ($\zeta = +0.122$, $f = 0.677$ Hz), i.e. *small-signal unstable*, whereas the meshed IEEE 14-bus stays stable even at 20× internal reactance or 0.02× inertia. The instability of the two-area case is therefore **topological** — it comes from the weak inter-area tie, not from low inertia or coupling strength, which a meshed grid tolerates.

6 Time-domain switching: SG trip and reclose

The demonstration run applies, over a 50 s window: a temporary three-phase fault $Z_f = j0.1$ pu at bus 4 during 1.0–1.15 s, an **SG trip at** 6 **s**, and a synchronised **SG reclose at** 9 **s**. The two disturbances are deliberately *well separated*: the fault is cleared 4.85 s before the breaker opens, so the post-fault transient has fully decayed and the IBR response to the trip is not contaminated by the fault. (This differs from the shorter 6 s objective run used for weight tuning in §9, which trips at 1 s and recloses at 4 s.) Figures 4–8 show the AGSI++ per IBR, the bus voltages, the frequencies, and the active-power injections. The ordered timeline is:

1. $[0, 1)$ s — SG is the slack; all four IBRs run as GFL at their scheduled dispatch.
2. 1.0–1.15 s — temporary fault $Z_f = j0.1$ pu; the IBRs *ride through* it as GFL (AGSI spikes but the fault clears and no IBR reaches the dwell-confirmed Γ_{on} crossing).
3. 1.15–6 s — fault-free recovery; the trajectory settles back to the SG-slack operating point before the next event, so the trip is a clean, independent disturbance.
4. $t = 6$ s — **SG trips**. All four IBRs cross Γ_{on} (AGSI $\approx$ 1.88–2.07) and **form GFM** after the $T_{d,on} = 0.5$ s dwell, i.e. at $t = 6.5$ s, so the island survives (network minimum voltage $\approx$ 0.49–0.53 pu on the formed island; min $|V| = 0.453$ pu at the trip transient).
5. $t = 9$ s — **synchronised reclose**; the angle reference is handed back to the SG and all four IBRs revert to their scheduled GFL dispatch. The reclose transient briefly re-triggers forming at $t = 9.5$ s, and AGSI++ peaks bounded at ≈ 3.5.
6. $t \to 50$ s — the response *stays bounded*: voltages recover ($V_{end} \approx 0.941$ pu at the weakest bus, higher elsewhere), the frequency excursion peaks near 61.4 Hz and then settles at $\approx$ 59.6 Hz (primary droop is present, but there is no secondary/integral frequency restoration to return exactly to 60 Hz), the SG re-takes $\approx$ 2.53 pu, and three of the four IBRs revert to GFL as their index falls below Γ_{off} at $t \approx$ 22.3–23.1 s (IBR2 at bus 3 remains GFM, supporting the weakest node). AGSI++ stays bounded ($\approx$ 3.5 peak) with no chatter.

Table 2: Main-run transition audit. "All" is an observed outcome of this case, not a group-switch rule.

Time	Event	Index/timer	Committed transition
0–1.0 s	no event	low AGSI++; timers idle	all remain GFL
1.0–1.15 s	temporary fault	above-line excursion shorter than $T_{d,on}$ = 0.5 s	GFL→GFL
6.0 s	SG trip	AGSI++ $\approx$ 1.88–2.07; per-IBR up-timers start	remain GFL during dwell
6.5 s	on-dwell matures	all local up-conditions persist	all GFL→GFM, index-driven
9.0 s	SG reclose	no down-threshold wait	all GFM→GFL, handback override
9.5 s	reclose transient	up-condition again matures	all GFL→GFM, index-driven
22.3–23.1 s	recovery	three down-conditions persist for $T_{d,off}$ = 1.0 s	three GFM→GFL; bus 3 remains GFM

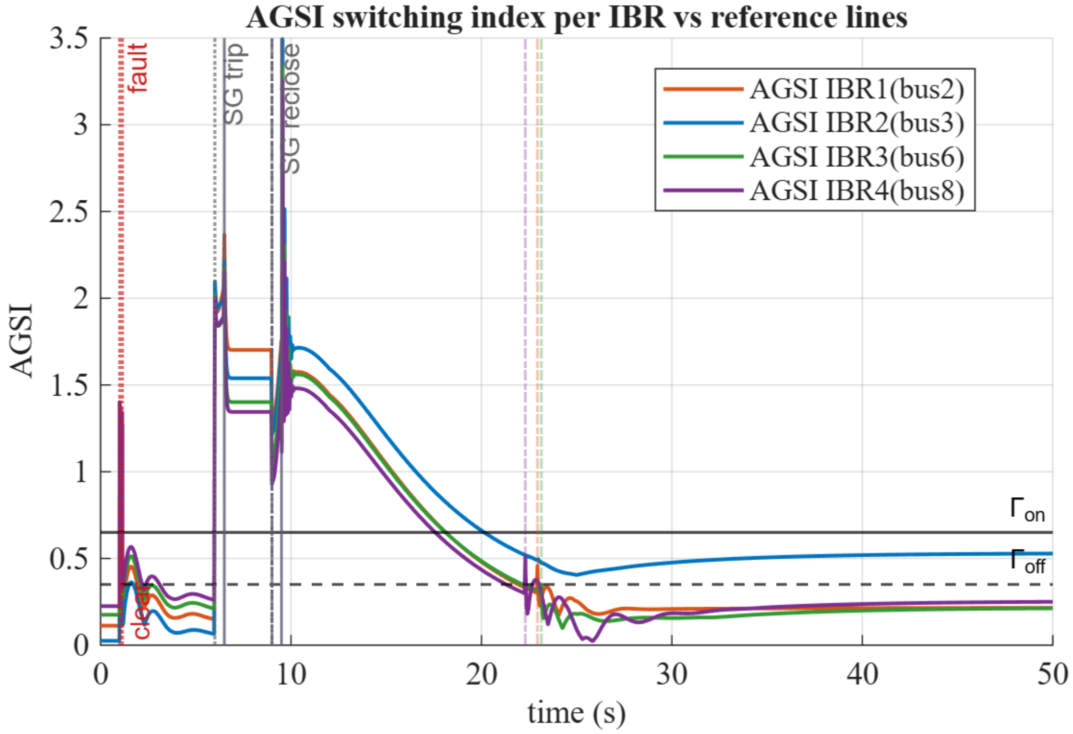

Figure 4: IEEE 14-bus: AGSI++ per IBR versus $\Gamma_{on} = 0.65$ / $\Gamma_{off} = 0.35$ (fault $Z_f = j0.1$ pu at 1.0–1.15 s, SG trip 6 s, reclose 9 s; 50 s window).

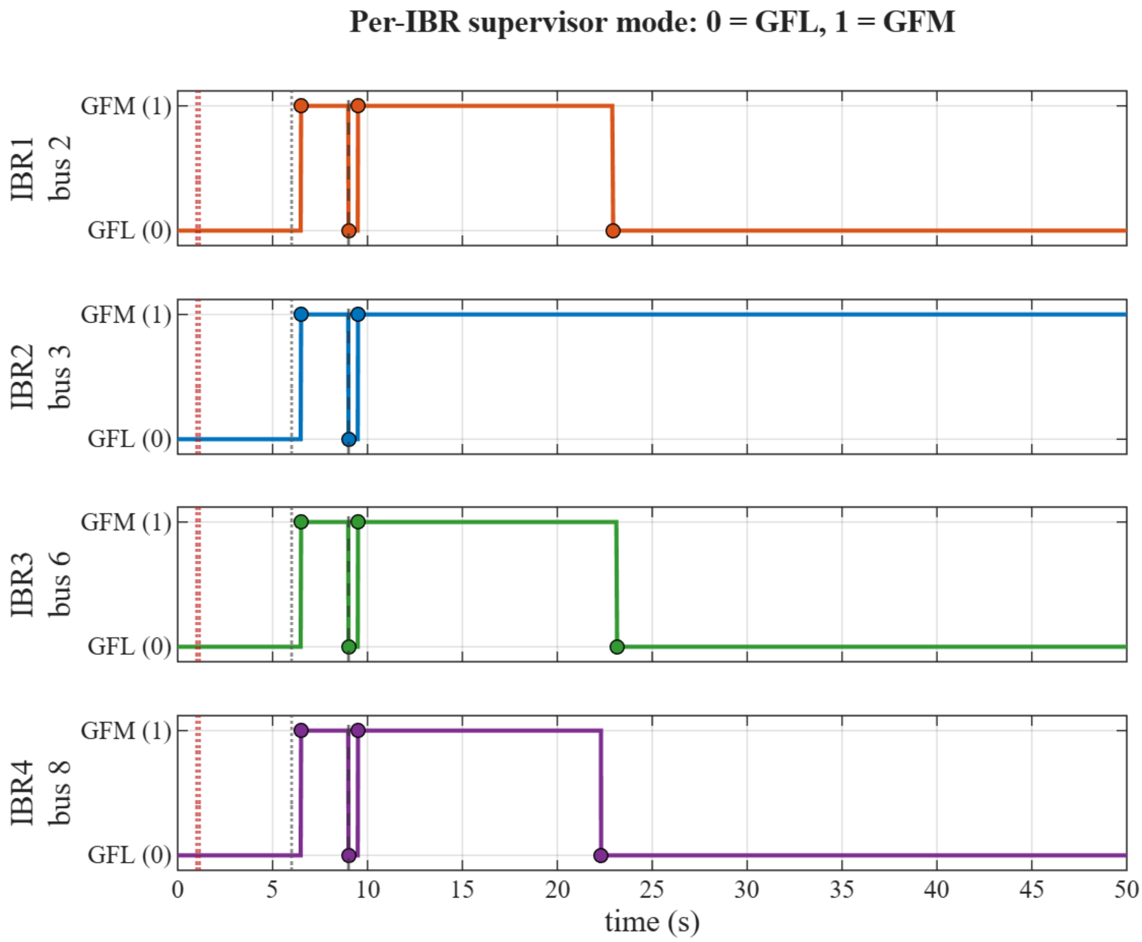

Figure 5: Recorded supervisor mode per device: 0 =GFL and 1 =GFM. Markers are committed transitions; the SG handback at 9 s is an event override, while the other transitions are per-device AGSI++/hysteresis/dwell decisions.

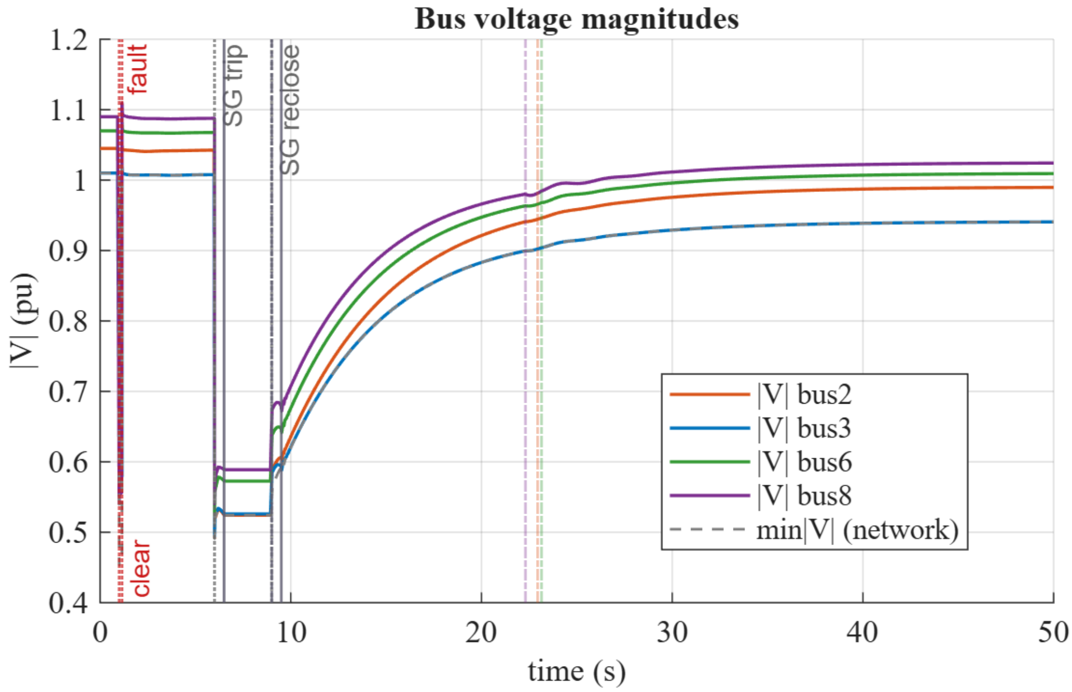

Figure 6: IEEE 14-bus: bus-voltage magnitudes through the fault / trip / reclose cycle; the network minimum recovers to ≈ 0.94 pu.

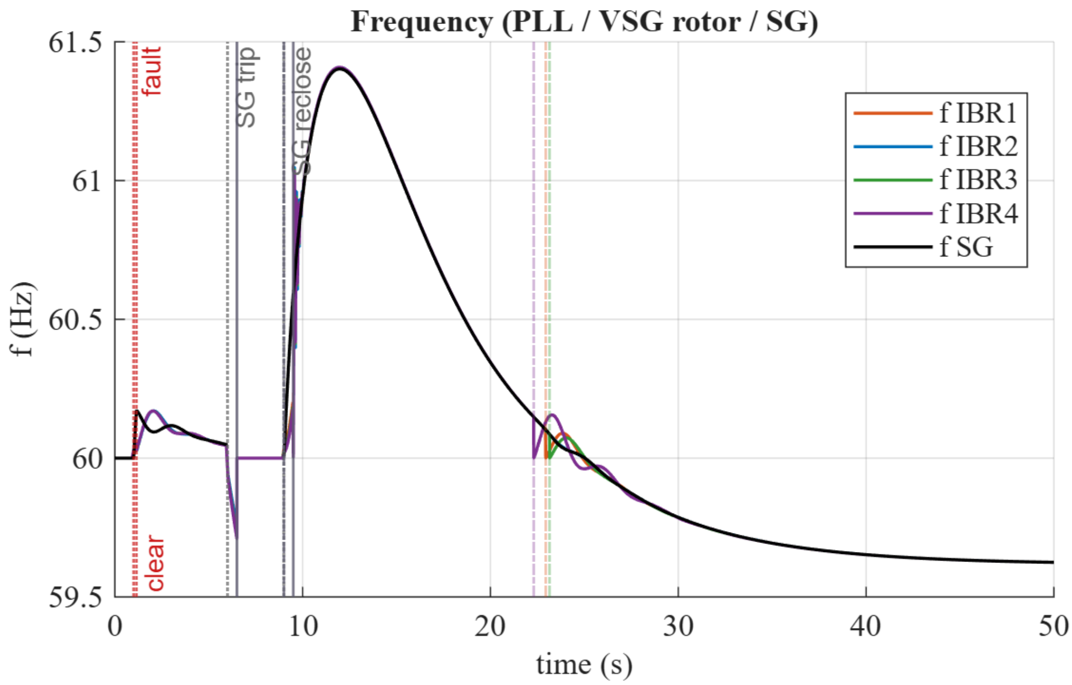

Figure 7: IEEE 14-bus: SG and IBR (PLL / VSG) frequencies; the post-trip excursion peaks near 61.4 Hz and then settles.

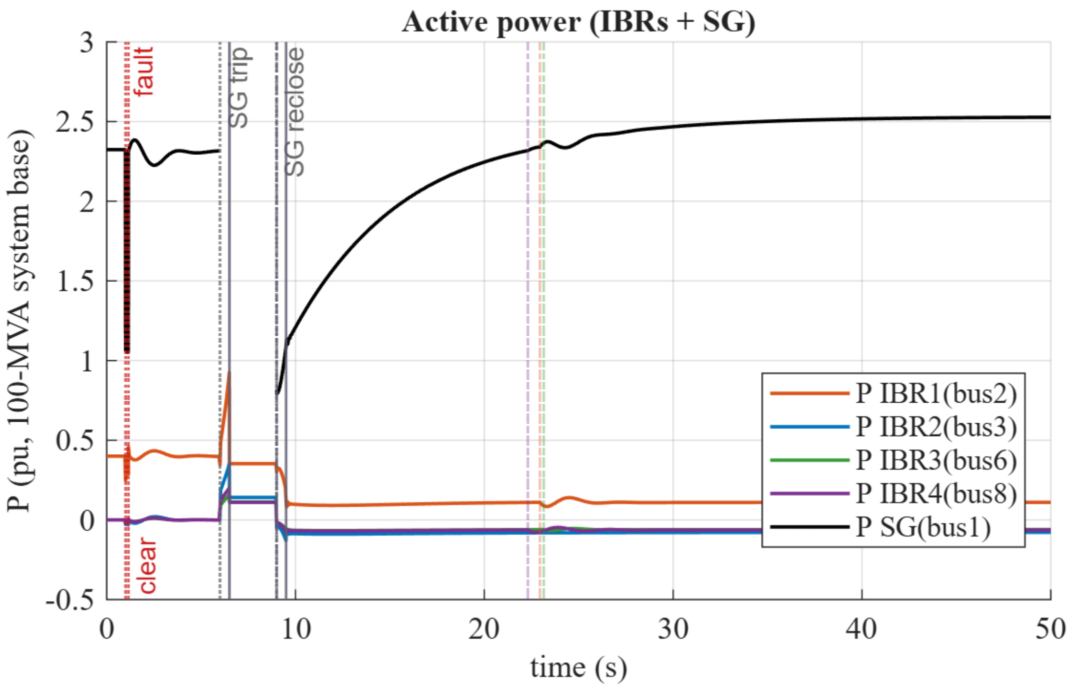

Figure 8: IEEE 14-bus: active-power injections. Pre-event the SG holds ≈ 2.32 pu and IBR1 (bus 2) holds ≈ 0.40 pu; after reclose the SG re-takes ≈ 2.5 pu.

7 Fault switching-outcome map

The switching map is a fault-only sweep (*no* SG trip, $T = 4$ s): a temporary three-phase shunt fault during 1.0–1.15 s is swept over fault reactance and fault bus,

$$X_f \in \{0.02, 0.05, 0.10, 0.20, 0.30, 0.50\} \text{ pu}, \qquad \text{fault bus} \in \{3, 4, 5, 9\},$$

and each run is classified into one of four outcome regions (Fig. 9, Table 3): *ride-through* (stayed GFL, no switch), *index-switch* (AGSI++ crosses Γ_{on} so ≥ 1 IBR forms GFM then reverts), *diverge / no post-clear recovery* (the run diverged, or the minimum voltage *measured after the fault is cleared* stays below the 0.25 pu collapse floor), and *non-converged* (fail-closed: the corrector did not converge, so no physical outcome is claimed). The collapse test deliberately uses the **post-clear** minimum voltage: while a low-impedance shunt fault is applied the faulted bus is pulled down *by construction*, so the during-fault sag carries no information about survival. Over the 24 fault points the tally is **ride-through** = 2**, index-switch** = 22**, diverge** = 0**, non-converged** = 0: every swept fault is cleared successfully and the network recovers to 0.98–1.01 pu, with the AGSI++ supervisor forming GFM on all but the two mildest points.

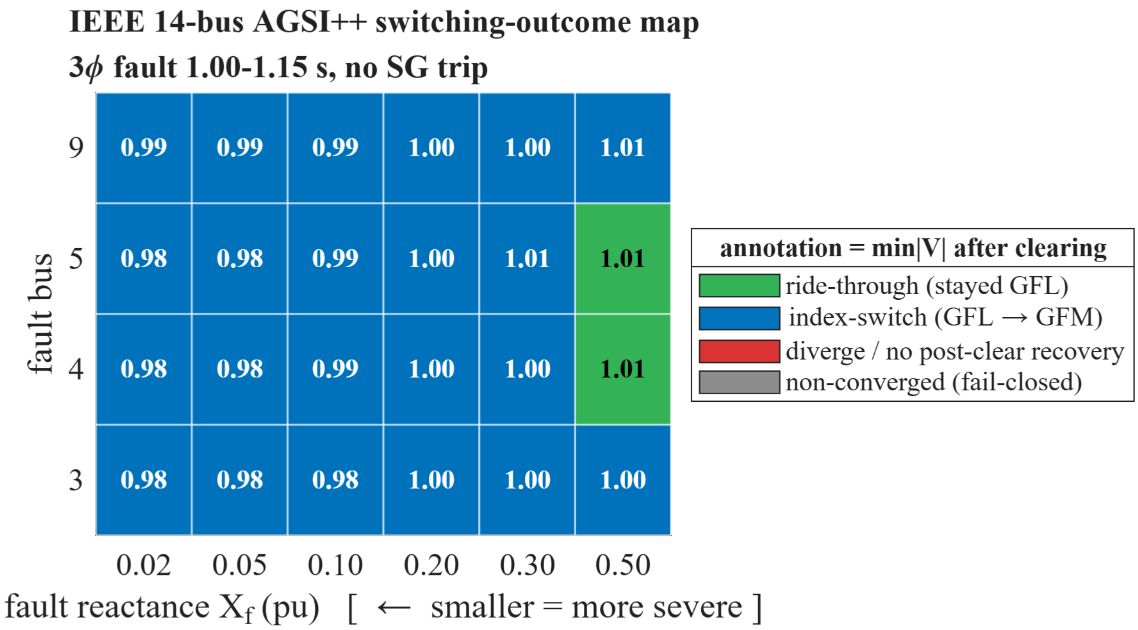

Figure 9: IEEE 14-bus AGSI++ switching-outcome map: fault reactance X_f (x, smaller = more severe) versus fault bus (y); colour = outcome class, annotation = minimum $|V|$ (pu) *after* the fault is cleared (the criterion used by the collapse test).

Table 3: Fault switching-outcome sweep (top, 24 points) and load-step sweep (bottom, permanent step at bus 4). Generated by `ieee14_switching_map.m`.

X_f (pu)	Fault bus	Outcome	min $\|V\|$ (pu) (incl. fault)	min $\|V\|_{\text{post}}$ (pu) (after clearing)	Total switches	Newton conv.
0.02	3	index-switch	0.102	0.980	8	yes
0.05	3	index-switch	0.224	0.985	8	yes
0.10	3	index-switch	0.371	0.983	10	yes
0.20	3	index-switch	0.549	1.001	4	yes
0.30	3	index-switch	0.649	1.002	6	yes
0.50	3	index-switch	0.759	1.004	4	yes
0.02	4	index-switch	0.136	0.979	8	yes
0.05	4	index-switch	0.287	0.978	9	yes
0.10	4	index-switch	0.450	0.987	10	yes
0.20	4	index-switch	0.628	0.995	7	yes
0.30	4	index-switch	0.723	0.999	10	yes
0.50	4	ride-through	0.819	1.009	0	yes
0.02	5	index-switch	0.143	0.979	8	yes
0.05	5	index-switch	0.298	0.982	8	yes
0.10	5	index-switch	0.463	0.987	8	yes
0.20	5	index-switch	0.639	0.995	7	yes
0.30	5	index-switch	0.734	1.006	4	yes
0.50	5	ride-through	0.827	1.009	0	yes
0.02	9	index-switch	0.075	0.990	10	yes
0.05	9	index-switch	0.169	0.988	10	yes
0.10	9	index-switch	0.293	0.992	8	yes
0.20	9	index-switch	0.462	0.998	8	yes
0.30	9	index-switch	0.571	1.004	4	yes
0.50	9	index-switch	0.701	1.007	2	yes

Load step	Step bus	Outcome	min $\|V\|$ (pu)	min $\|V\|_{\text{post}}$ (pu)	Total switches	Newton conv.
+10%	4	ride-through	1.008	1.008	0	yes
+30%	4	ride-through	1.003	1.003	0	yes
+50%	4	ride-through	0.999	0.999	0	yes

Physics of the map. Once the collapse test is applied where it belongs — after clearing — a clean *switching* gradient emerges rather than a collapse region. Every one of the 24 faults is survived: the deepest during-fault sags reach min $|V| = 0.075$–0.14 pu (bolted-ish $X_f = 0.02$), yet after the breaker clears at 1.15 s the network recovers to 0.98–0.99 pu, so no point is a voltage collapse and every Newton step converged. What the severity actually decides is *whether the*

AGSI++ supervisor commands forming: from $X_f = 0.02$ up to 0.30 at every bus (and up to 0.50 at IBR bus 3 and the electrically weak bus 9) the sag pushes AGSI++ over Γ_{on}, so at least one IBR forms GFM and then reverts ⇒ *index-switch* (22 points, 2–10 transitions each). Only the two mildest cases ($X_f = 0.50$ at the stiffer load buses 4 and 5) stay GFL with zero switches ⇒ *ride-through*. Fault location matters as expected: at the IBR bus 3 and the weak bus 9 even the mildest X_f triggers switching, whereas the stiffer load buses 4/5 do not. The load-step sub-table confirms the interpretation: permanent +10/30/50% load steps at bus 4 (no SG trip) all ride through with $\min|V| \geq 0.999$ pu and zero switches — a load step is far milder than a fault and never drives an IBR to form.

8 Event coverage and present model limits

The existing evidence covers: no-event hold (no false switch); mild cleared faults (ride-through); stronger/weak-bus faults (22 of 24 points index-switch and return, with all 24 recovering); permanent +10/30/50% load steps at bus 4 (no switch); SG trip (index-driven forming); and SG reclose (coordinated handback followed by index-driven re-forming/recovery). A switch is not required for a test to pass: retaining GFL under a disturbance that never satisfies threshold and dwell is the correct supervisor outcome.

The claims are bounded by the current model: positive-sequence RMS and balanced three-phase shunt faults, not EMT/PWM/harmonic or unbalanced-fault simulation; reduced-6 IBR branches without a full DC-link, LCL, SOC, thermal, vendor-protection or LVRT/anti-windup stack; a study-level 1.2-rated current-magnitude limiter; a manual-field 4-state SG (no AVR/PSS, primary droop but no secondary frequency restoration); and constant-impedance dynamic loads. The scheduled reclose is made synchronous by re-initialisation and does not model a synchronism-check relay or failed close. Bumpless transfer proves $I^+ = I^-$ only at the switching instant, not zero subsequent voltage, frequency or power transient. AGSI++ weights, thresholds and dwells are frozen diagnostic study choices, not certified protection settings; online BO/MPC, measurement noise/latency and broad parameter/topology uncertainty are not covered. Non-converged trajectories fail closed and support no physical claim.

These statements were checked against the runtime mode/timer/transfer implementation in `SwitchableIbr6`, the accepted-step and coordinated-reclose path in `padiyar_switch_tds`, the IEEE14 builder, and the no-event/fault/bidirectional/hysteresis tests.

9 AGSI++ weight-tuning provenance

The hand-set AGSI++ weights and thresholds were checked against an in-house, offline, design-time search (`scripts/reporting/optimize_agsi_weights.m`; project-owned base MATLAB, **no external optimization solver**, not on any production runtime path). Guided by the EECON49-P4 Bayesian-Optimization *concept*, the search is a uniform-simplex random sampling of the six weights (each ≥ 0, sum $= 1$) plus Γ_{on}/Γ_{off}, followed by a coordinate local refine, minimising, for the IEEE 14-bus SG-trip@1 s + reclose@4 s run over $T = 6$ s:

$$\begin{aligned} J = 12 \cdot \overline{|1 - V_{\min}(t)|} \ + \ 0.6 \cdot (\text{total mode switches}) \\ + \ 0.05 \cdot \max \text{AGSI} \quad (+\,10^5 \text{ penalty on divergence}). \end{aligned} \tag{5}$$

The search improves the objective by only 0.4% (J : 11.959 → 11.906), i.e. the hand-set weights and thresholds are *already near-optimal* for this stable operating point. The optimum leans toward the grid-strength term (w_{SCR}) and drops w_{lock}, consistent with weak-grid GFL instability being the dominant stress, but the gain is negligible and the hand-set values are more physically interpretable and balanced. **Decision:** keep the frozen defaults $[w_V, w_f, w_R, w_P, w_{SCR}, w_{lock}] = [0.25, 0.25, 0.15, 0.10, 0.15, 0.10]$, $\Gamma_{on} = 0.65$, $\Gamma_{off} = 0.35$; the search is retained only as a reproducible tool that *validates* the choice (`ASSUMED_DIAGNOSTIC`; frozen before use).

10 Summary

On the meshed IEEE 14-bus system, one manual (no-AVR) synchronous generator and four GFL IBRs form a small-signal-stable operating point (28 states, zero RHP eigenvalues, least-damped 0.345 Hz at $\zeta = +0.267$) that stays stable across the whole inertia and coupling sweep; the AGSI++ supervisor rides through a temporary fault as GFL, forms GFM on the SG trip to hold the island, and hands the reference back on reclose while staying bounded and chatter-free. The switching-outcome map shows that all 24 swept faults are cleared and recovered (no collapse point), with fault severity and location deciding only whether AGSI++ commands forming (22 index-switch, 2 ride-through), and the offline weight search confirms the hand-set AGSI++ weights are near-optimal and frozen. The stark contrast with the small-signal-unstable Padiyar two-area case (max Re = +0.058 s^{-1}) shows the difference is topological.

References

[1] EECON49-P4 (KMITL), "A Bayesian Optimization-Based GFL/GFM Mode Switching Strategy," The 49th Electrical Engineering Conference (EECON-49), 2026.

[2] H. Ding *et al.*, "A novel design for switchable grid-following and grid-forming control," *IEEE Trans. Sustain. Energy*, vol. 16, no. 2, pp. 1301–1314, 2025.

[3] P. Liu, X. Xie, and J. Shair, "Adaptive hybrid grid-forming and grid-following control of IBRs under varying SCRs," *IEEE Trans. Power Electron.*, vol. 39, no. 6, pp. 6603–6607, 2024.

[4] X. Gao *et al.*, "Seamless switching method between grid-following and grid-forming control," *IEEE Trans. Ind. Appl.*, vol. 61, no. 1, pp. 597–606, 2025.

[5] ENTSO-E, "Rate of Change of Frequency (RoCoF) withstand capability," Network Code RfG implementation guidance document, 2018.